

**EBN122 EKSAMEN/EXAMINATION (14 NOV 2014)**

**ANTWOORDBLAD / ANSWER SHEET**

|                         |      |                     |  |  |  |  |  |  |  |
|-------------------------|------|---------------------|--|--|--|--|--|--|--|
| Van / Surname:          | Memo | Student #:          |  |  |  |  |  |  |  |
| Voorletters / Initials: |      | Tel. nr. / Tel no.: |  |  |  |  |  |  |  |

|    |    |    |    |    |    |    |    |      |     |
|----|----|----|----|----|----|----|----|------|-----|
| Q1 | 15 | Q2 | 30 | Q3 | 22 | Q4 | 23 | Tot: | /90 |
|----|----|----|----|----|----|----|----|------|-----|

**VRAAG 1 – QUESTION 1 (15)**

|     |                                             |                         |                                             |
|-----|---------------------------------------------|-------------------------|---------------------------------------------|
| 1.1 | a) $W = 0,125 \text{ kWh}$ //               |                         | b) $Q = -7.5 \mu\text{C}$ //                |
| 1.2 | $P_1 = 48 \text{ W}$ //                     | $P_2 = 30 \text{ W}$ // | $P_3 = 320 \text{ W}$ //                    |
| 1.3 | $I_S = 10 \text{ A}$ //                     |                         | $v_x = -12 \text{ V}$ //                    |
| 1.4 | $R_{ab} \text{ (Circuit 1)} = 33 \Omega$ // |                         | $R_{ab} \text{ (Circuit 2)} = 30 \Omega$ // |

**VRAAG 2 – QUESTION 2 (30)**

|     |                                       |             |                                       |              |
|-----|---------------------------------------|-------------|---------------------------------------|--------------|
| 2.1 | $a = 11$ //                           | $b = -5$ // | $c = -5$ //                           | $d = 7$ //   |
| 2.2 | $e = 13k$ //                          | $f = 9k$ // | $A = i_2$ //                          | $B = i_1$ // |
| 2.3 | a) $R_B = 430k\Omega$ //              |             | b) $V_{CE} = 5V$ //                   |              |
| 2.4 | $V_0 = -40V$ //                       |             |                                       |              |
| 2.5 | $V_{ab} = -2.4mA$ //                  |             |                                       |              |
| 2.6 | $V_{cd} = 1.5V$ //                    |             |                                       |              |
| 2.7 | $V_{TH} = 22.5V$ //                   |             | $R_{TH} = 6\Omega$ //                 |              |
| 2.8 | $I_N = 0,625A$ //                     |             | $R_N = 20\Omega$ //                   |              |
| 2.9 | a) $P_{max}(RL=5\Omega) = 0,3125W$ // |             | a) $P_{max}(RL=1\Omega) = 0,2778W$ // |              |

**VRAAG 3 – QUESTION 3 (22)**

|     |                                      |                                         |                |
|-----|--------------------------------------|-----------------------------------------|----------------|
| 3.1 | $v_{0(R=60k\Omega)} = 4V$ ✓✓         | $i_{0(R=\infty)} = -0.44 \text{ mA}$ ✓✓ |                |
| 3.2 | $i_1 = -2 \text{ mA}$ ✓✓             | $i_2 = 0.15 \text{ mA}$ ✓✓              |                |
| 3.3 | $v_I = 6.5V$                         | $v_{a \min} = -2V$ ✓✓                   |                |
| 3.4 | $V_1/V_{s1} = -4$ ✓✓                 | $V_2/V_{s2} = 2.25$ ✓✓                  | $V_3 = -2V$ ✓✓ |
| 3.4 | $W_C = 6 \text{ mJ}$ ✓✓              |                                         |                |
| 3.5 | $i(1\text{ms}) = -3.83 \text{ A}$ ✓✓ |                                         |                |



VRAAG 4 – QUESTION 4 (23)

|     |                                                          |                                                        |
|-----|----------------------------------------------------------|--------------------------------------------------------|
| 4.1 | $\theta = 90$ ✓✓ (degrees)                               | $C = 2$ $\mu F$ ✓✓                                     |
| 4.2 | $Z_{ab} = j34.29 \Omega$ ✓✓ (rectangular form)           |                                                        |
| 4.3 | $\bar{V} = 353.5 \angle -15^\circ V$ ✓✓ (polar form)     | $\bar{I} = 10.92 \angle 139.1^\circ A$ ✓✓ (polar form) |
| 4.4 | $\bar{I}_N = 0.4 \angle -60^\circ A$ ✓✓ (polar form)     | $Z_N = 200 + j100 \Omega$ ✓✓ (rectangular form)        |
| 4.5 | $\bar{V}_{Th} = 60 \angle -36.9^\circ V$ ✓✓ (polar form) | $Z_{Th} = 8.64 + j11.52 \Omega$ ✓✓ (rectangular form)  |
| 4.6 | $\bar{V}_0 = 75 \angle 0^\circ V$ ✓✓ (polar form)        |                                                        |
| 4.7 | $A \angle \theta^\circ = 2 \angle 90^\circ$ ✓            |                                                        |
|     | $a_1 + jb_1 = 20 - j20 \Omega$ ✓                         | $a_2 + jb_2 = 30 + j25 \Omega$ ✓                       |